A QUANTITATIVE LOWER BOUND FOR THE BOUNDARY FUNCTION OVER THE WHOLE RANGE OF SET SIZES

SAHAR DISKIN, ANNA GEISLER

Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The boundary function f-star of the section and the two estimates on it coming from the expansion inequalities. Set up in the author's section on the random graph process, with the author's displays labelled eq:expansion and eq:f-star (source lines 795-810). Supplies the two lower bounds for the boundary function that the proof evaluates in each range of the size parameter.

Source reference. Original equation 6.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 807–807. Navigation pointer only: the counted proof uses the author's displayed expansion inequality as one of its two inputs; the display is not reproduced in this card.

Source reference. Original equation 6.1: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 801–801. Navigation pointer only: the counted proof uses the author's displayed two-term estimate on the boundary function; the display is not reproduced in this card.

Lemma 0.1. *For all sufficiently large d and $3 \le k \le 2u_1$,*

$$f^*(k) \ge 2d + 15(M-1)k\ln d. \tag{1}$$

Proof. For $k = 3$, (6.2) together with the first term of (6.1) give

$$f^*(k) \ge 3d - 3(M-1)\log_M(3) \ge 2d + 45(M-1)\ln d,$$

as claimed. For $4 \le k \le d^3$ the same implies

$$f^*(k) \ge kd - 3k(M-1)\log_M(d) \ge 3d + 15k(M-1)\ln d.$$

If $d^3 < k \le 2u_1$, then (6.2) together with the second term of (6.1) give

$$f^*(k) \ge \frac{ek}{M}\left(\frac{M^3}{p_-}\ln d - \ln 2\right).$$

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Using that $p_- \leq 1 - 1/M$, we have $\frac{eM^2}{p_-} \geq 16(M-1)+1$. This yields

$$f^*(k) \geq 16(M-1)k\ln d + k\ln d - \frac{ek}{M}\ln 2 \geq 3d + 15(M-1)k\ln d,$$

where the last inequality uses that $k > d^3$. □